VATSAL THAKOR

+91 9978407849 ⋄ Ahmedabad, India ⋄ LinkedIn ⋄ GitHub

EDUCATION

B.Tech in Electronics and Communication Engineering
Nirma University, Ahmedabad
Expected May 2026 — CGPA: 7.71/10

Relevant coursework: Electronic Devices and Circuits, VLSI Design, Analog CMOS IC Design, Testing and Verification of Digital Circuits, FPGA-Based System Design, Digital Logic Design, Analog Electronics, Machine Learning

RESEARCH INTERESTS

TCAD Process Modeling, Semiconductor Device Physics, Plasma Etching Simulation, Advanced Memory Devices (3D NAND), VLSI Design

SKILLS

Hardware Description Languages: Verilog HDL, SystemVerilog
Programming: Python, C, Embedded C, Assembly
EDA Tools: Cadence Virtuoso, Synopsys Sentaurus Topography 3D
FPGA Tools: Xilinx Vivado, Quartus, ModelSim
Technical Software: MATLAB, Microwind, Keil uVision
Mechanical CAD: SolidWorks, Creo Parametric

RESEARCH EXPERIENCE

Research Intern December 2025 – April 2026
Indian Institute of Technology Gandhinagar (IITGN)

- Developed a 3D process simulation framework for plasma-based High Aspect Ratio (HAR) etching of ONO (Oxide–Nitride–Oxide) stacks used in 3D NAND memory structures.
- Implemented and calibrated $Ar/C_4F_8/O_2$ plasma chemistries in Synopsys Sentaurus Topography 3D to model polymer deposition, ion-enhanced etching, and radical transport.
- Analyzed radical angular distribution, reflection probability, reflection exponent, nucleation and growth probabilities to study polymer build-up, trench bowing, and distortion mechanisms.
- Optimized plasma process parameters to reduce polymer clogging at trench entrances and improve sidewall protection in high-aspect-ratio structures.
- Developed automated scripts to extract polymer thickness and volume metrics from Sentaurus TDR outputs for quantitative profile analysis.

PROJECTS

CMOS Operational Amplifier Design and Performance Analysis using Cadence Virtuoso

- Designed and simulated a CMOS operational amplifier in Cadence Virtuoso, analyzing key performance metrics, including Slew Rate, CMRR, PSRR, and GBW.

- Performed DC, AC, and transient simulations to extract and validate op-amp characteristics under various operating conditions.
- Optimized the transistor-level design for improved frequency response and stability, ensuring high performance and accurate analog behavior.

Raster-to-Vector Image Conversion using MATLAB Image Processing

- Developed a MATLAB-based image processing system for vectorizing raster images with adjustable parameters such as edge threshold, color quantization, and smoothing level.
- Implemented efficient matrix and vector operations to enhance processing speed and scalability for high-resolution images.
- Validated results through visualization and parameter tuning to achieve optimal edge accuracy and noise reduction across diverse image types.

FPGA-Based Morse Code Generator using Verilog HDL

- Designed a Morse code generator in Verilog to convert alphanumeric inputs into timed dot and dash sequences using a finite-state machine and lookup table.
- Simulated and verified functionality through testbenches before synthesizing and implementing it on an FPGA board.
- Validated timing accuracy and output through LED and buzzer signals for various input messages.

FSM-Based Secure Door Lock System with Three-Failure Alarm using Verilog HDL — GitHub

- Designed a password-based door lock system in Verilog using a finite-state machine for secure access control.
- Simulated and tested functionality in Verilog before deploying on FPGA, ensuring reliable response and reset operation.
- Implemented a three-failure intrusion alarm mechanism triggered after consecutive incorrect password attempts.

UART-Based Serial Communication between Two 8051 Microcontrollers — GitHub

- Developed a reliable serial communication protocol between two 8051 microcontrollers, enabling reliable UART-based data transmission and reception.
- Configured Timer 1 for baud rate generation, ensuring stable 9600 bps communication between transmitter and receiver microcontrollers.
- Implemented logic to compare received data and trigger specific hardware actions based on the input, demonstrating efficient data handling and control.